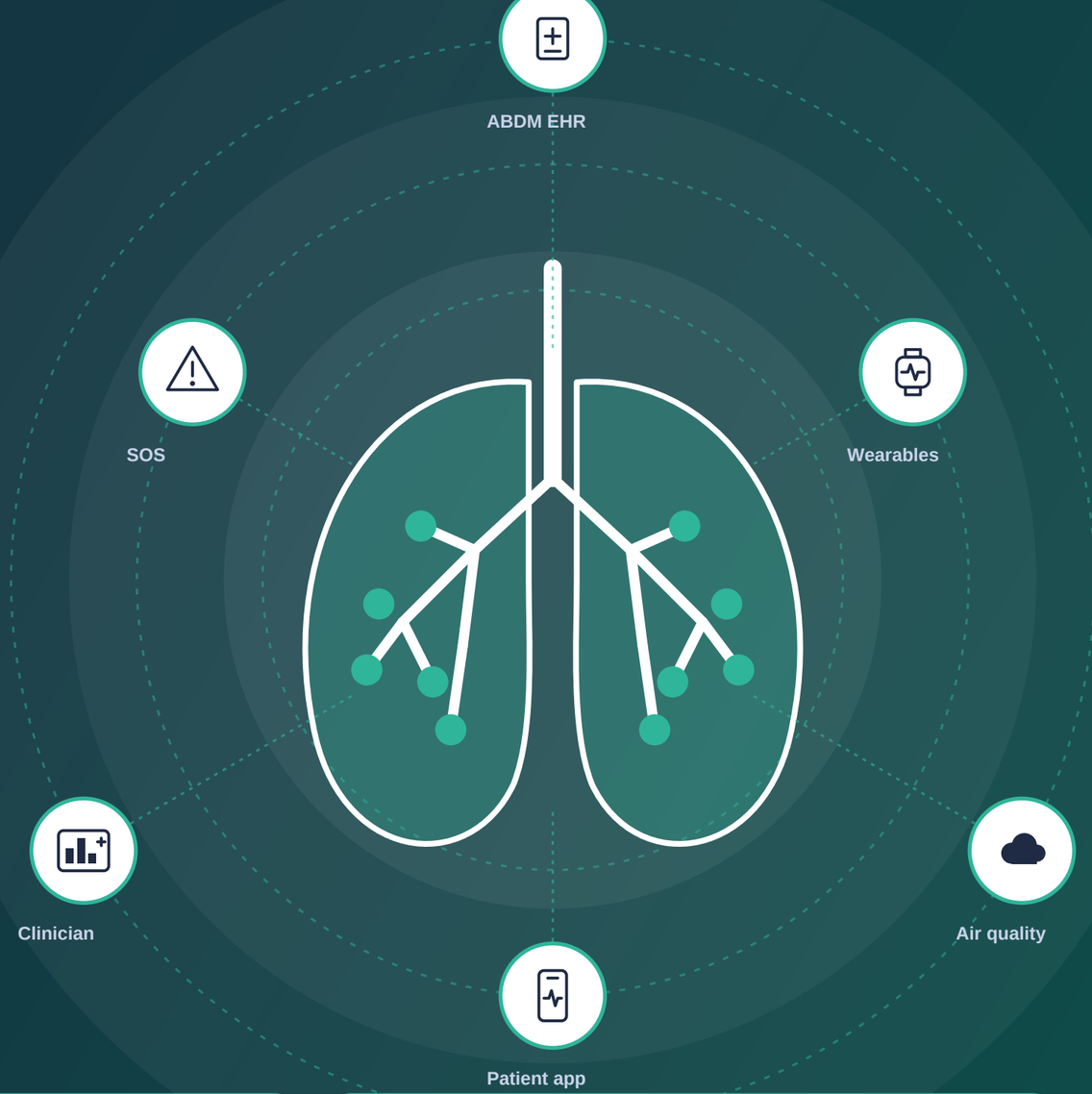
PulmoTwin

Multimodal Cardiopulmonary Digital Twin

for Proactive Adverse Event Forecasting and Personalised Clinical Intervention

Seeing an asthma attack coming, hours before it starts



SUBMITTED TO

Digital Twin Challenge 2026

Happiest Health | Reimagining & Reforming Healthcare in India Summit, Bengaluru

Team PulmoTwin

EXPLORE THE PROJECT (click to open)

GitHub repository
github.com/Kabirroy12345/Digital_Twin

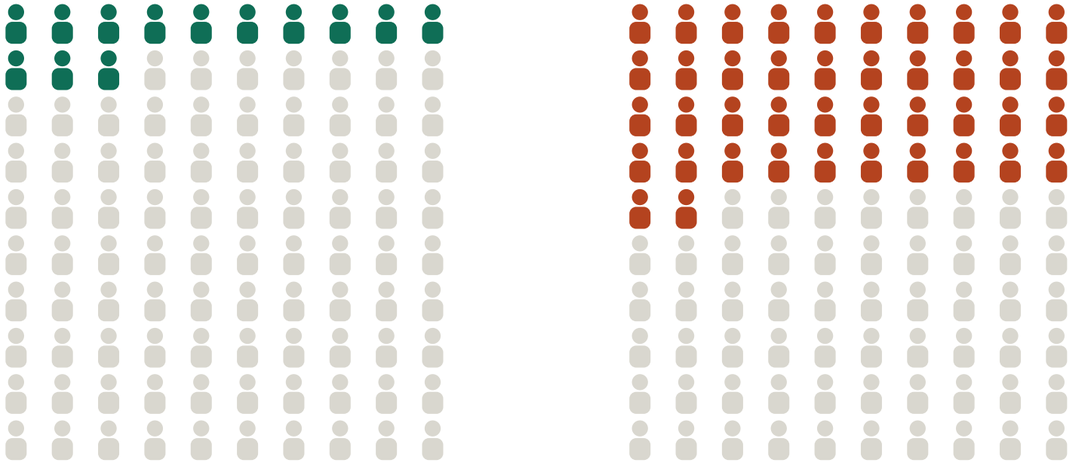
Live interactive demo
kabirroy12345.github.io/Digital_Twin

Full demo video
youtube.com/watch?v=i7NJstq8_ZE

Research prototype, not a medical device. Built on synthetic data. In an emergency dial 112 / 108. MIT license.

Background: why asthma, why India, why now

India, out of every 100 people in the world's asthma burden

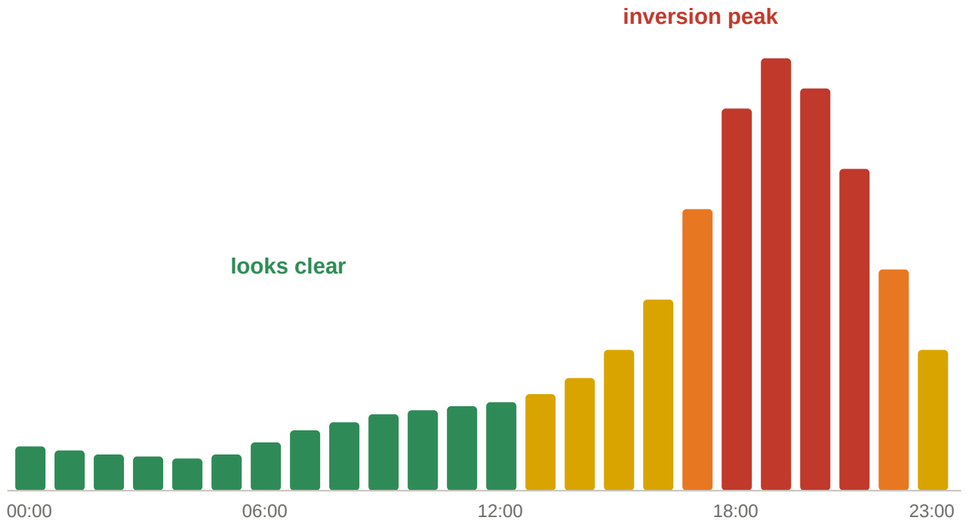


13% of the world's asthma patients live in India

42% of the world's asthma deaths happen in India

Each icon is 1 of 100. Source: Lancet Global Health, as cited in the project README. 37.9M Indians live with asthma.

A clear morning can hide a smoggy evening



Illustrative hourly PM2.5 pattern in a winter metro

WHY NOW: FOUR TRENDS CONVERGE

Smog seasons
Pollution spikes by the hour

ABDM records
National FHIR-based health records

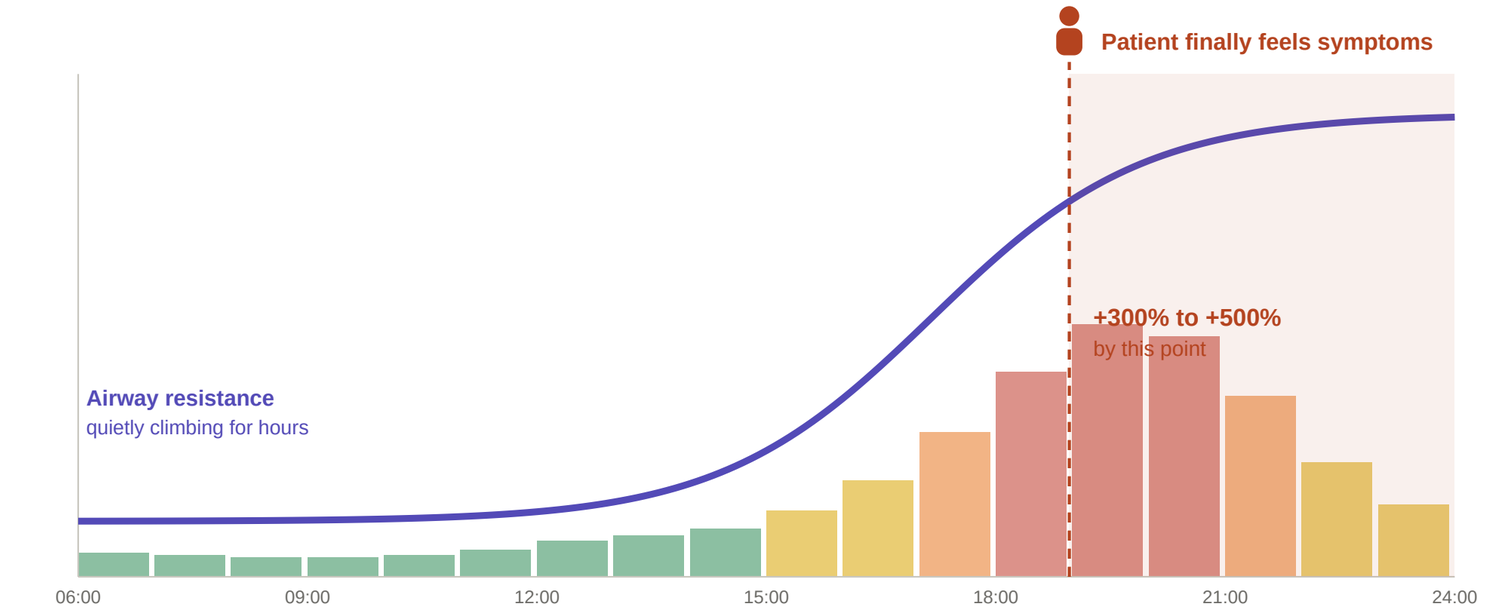
Wearables
Affordable sensors and smart inhalers

Digital twins
Idea from engineering reaches medicine

PulmoTwin

The problem today: asthma care starts too late

What happens to one patient on a smoggy evening



Bars: hourly PM2.5 (illustrative). Line: airway resistance. Shape is illustrative; the 300-500% rise is from the project README.

Care begins after symptoms

Patients check air quality only once they cough.

Nothing joins the signals

Record, wrist and weather site never meet for one person.

Doctors see a blur

Over 80% cannot recall a month of symptoms and puffs.

THE PATIENT JOURNEY TODAY

The patient only learns about the problem in the last two steps

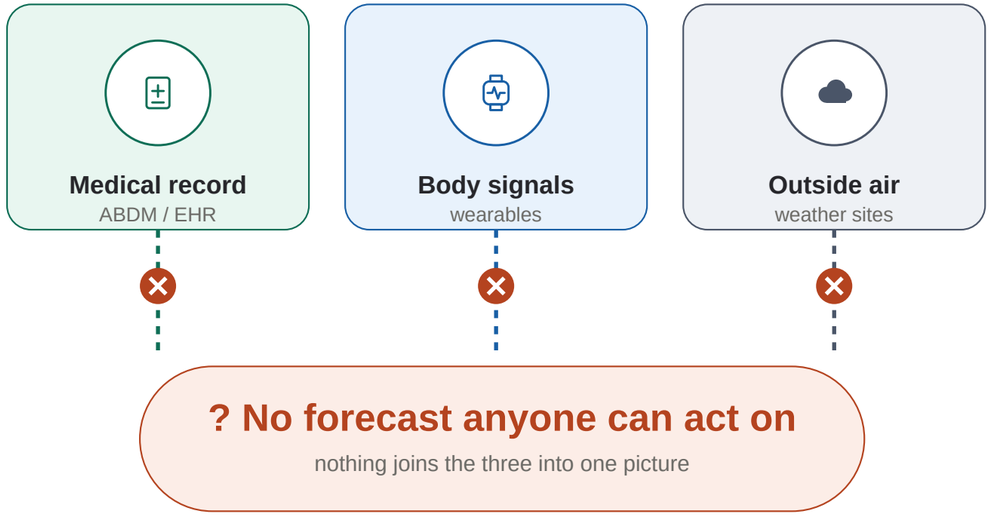
Feels fine → Air turns bad → Airways tighten → Symptoms appear → Rescue care

Problem statement

HOW MIGHT WE...

give people with asthma in India, and the doctors who treat them, an early and personal warning of an attack, hours before symptoms begin, along with a safe way to decide what to do about it?

THE RISK IS SPLIT ACROSS THREE SILOS



WHAT A GOOD ANSWER MUST DO

Predict early
Flag a surge 8 to 24 h before onset

Personalise
One patient, one physiology, one place

Guide the action
Show what works before trying it

Stay safe
Never diagnose, always escalate

CHALLENGE FIT

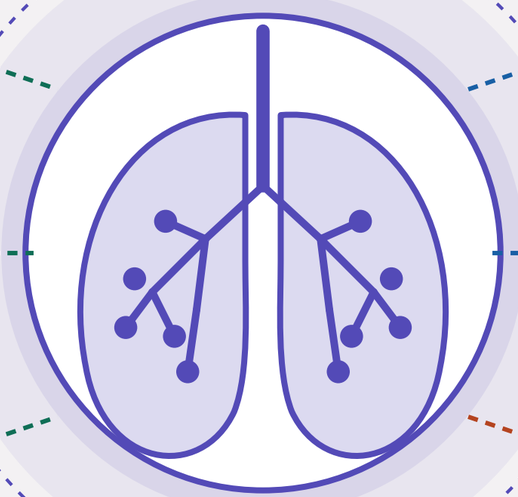
- Chronic condition common in India: asthma
- Static + dynamic data: EHR + wearables
- Predictive model: stacked ML
- Doctor dashboard: physician console

Our solution: a living virtual replica of the patient

Medical history
ABDM / FHIR R4 record

Body signals
Wearable + inhaler stream

Local air
Open-Meteo GPS feed



PulmoTwin
digital twin of one person's airways

Patient app
Safe window, plan, alerts

Physician console
Triage, sign-off, PDF

Emergency path
One tap to 108 / 112

See
Rebuilds lungs and stress response from live data

Foresee
Flags a surge 8 to 24 h early

Test
Simulates options before acting

Protect
Safety rails and human approval

It assists. It never diagnoses and never changes a prescription.

Implementation: how we built it

1 Build the data

Synthetic cohort, FHIR record connector, wearable stream, live air quality

```
data_generator.py
ehr_connector.py
wearable_sim.py
```

2 Model the physiology

Equations turn vitals into resistance, compliance, stress and a risk index

```
digital_twin/
twin_engine.py
```

3 Train the risk model

Stack LR, RF and GB; calibrate; test on 300 held-out cases

```
models.py
preprocess.py
```

4 Add what-if + agent

Compare interventions; plan the day around the air

```
counterfactual.py
agent/
```

5 Wrap in safety

Distress bypass, GINA rule, output filter, approvals, audit

```
guardrails
in agent + app.py
```

6 Ship and verify

Flask API, web UI, pytest, GitHub Pages, Render, Vercel, Docker

```
app.py
web_ui/ tests/
```

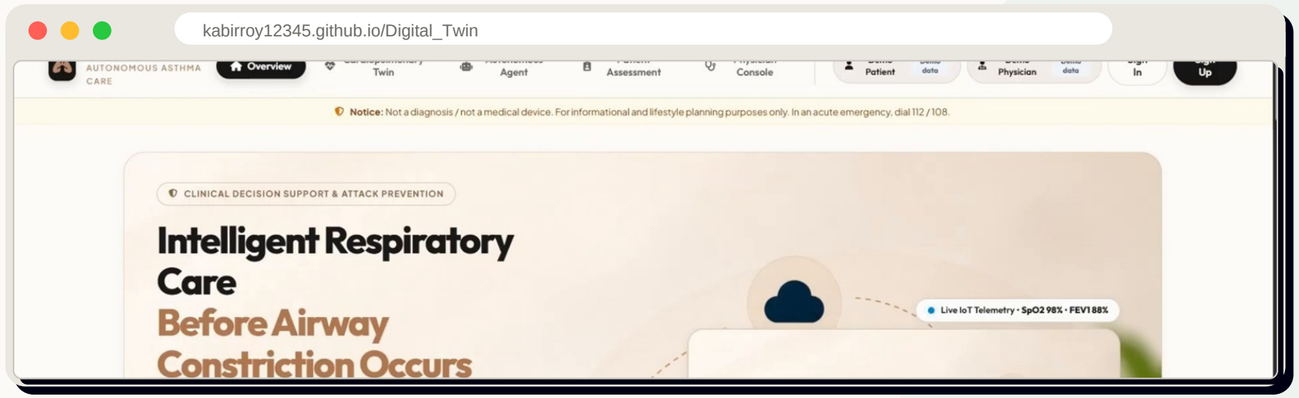
WHAT GOT BUILT

3 data streams

4 safety rails

5 app pages

2 languages



Technical aspects: the twin engine and the AI

Airway resistance



2.34

cmH2O/(L/s)

`Raw = 1.8 / PEF^1.6`

Lung compliance



54.3

mL/cmH2O

`Cdyn = 70*PEF - FeNO/10`

A-a O2 gradient



13.8

mmHg

`from SpO2, resp. rate`

Autonomic tone



0.27

relaxed

`from HRV rMSSD`

Risk index (ERI)



0.24

stable

`composite 0 to 1`

Values are the demo patient's. Gauge zones are illustrative.

Risk ensemble: from three models to one safe answer

Logistic Regression
weight 1

Random Forest
100 trees, weight 2

Gradient Boosting
150 stages, weight 2

SOFT VOTE

Entropy gate

low-confidence cases are rejected

GINA override

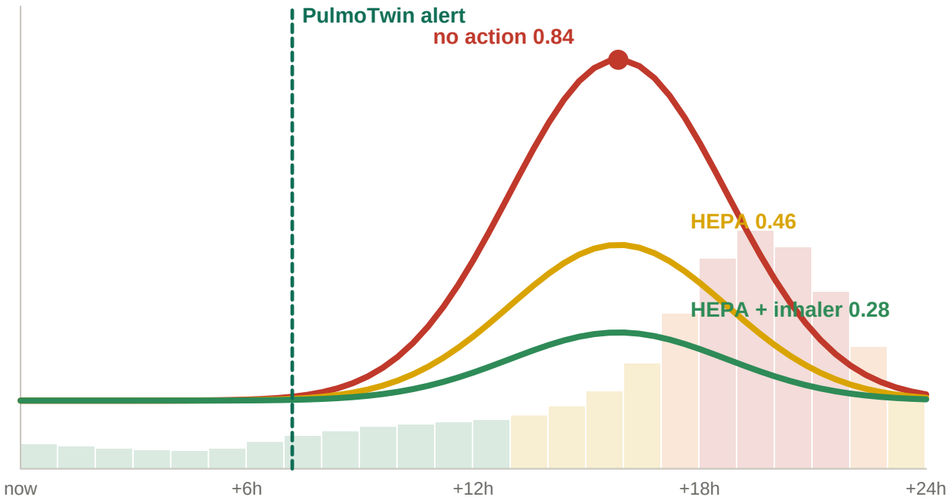
daily symptoms => High

LOW **MEDIUM** **HIGH**

Risk tier shown to patient and clinician

Soft vote averages probabilities of the three models

24-hour projection and what-if



Peaks come from the simulation; curve shape and alert time are illustrative.
Bars: forecast air-quality driving the model.

Technical aspects: the stack, data and quality

Python 3.11
twin engine, ML, API

Flask 3.1
REST API, Gunicorn

SQLite
storage + audit trail

NumPy + SciPy
ODE kinetics

scikit-learn
LR, RF, GB stacking

ReportLab
30-day PDF reports

HL7 FHIR R4
EHR schema

ABDM
national health IDs

Open-Meteo
air-quality + weather

Docker
container image

HTML5 + ES6 JS
web cockpit

GitHub Pages
live demo hosting

Marks are simplified vector drawings, not official brand logos.

Data and privacy

Synthetic data only (Synthea and MIMIC-IV distributions), zero personal information. Export and erase aligned with DPDP Act 2023.

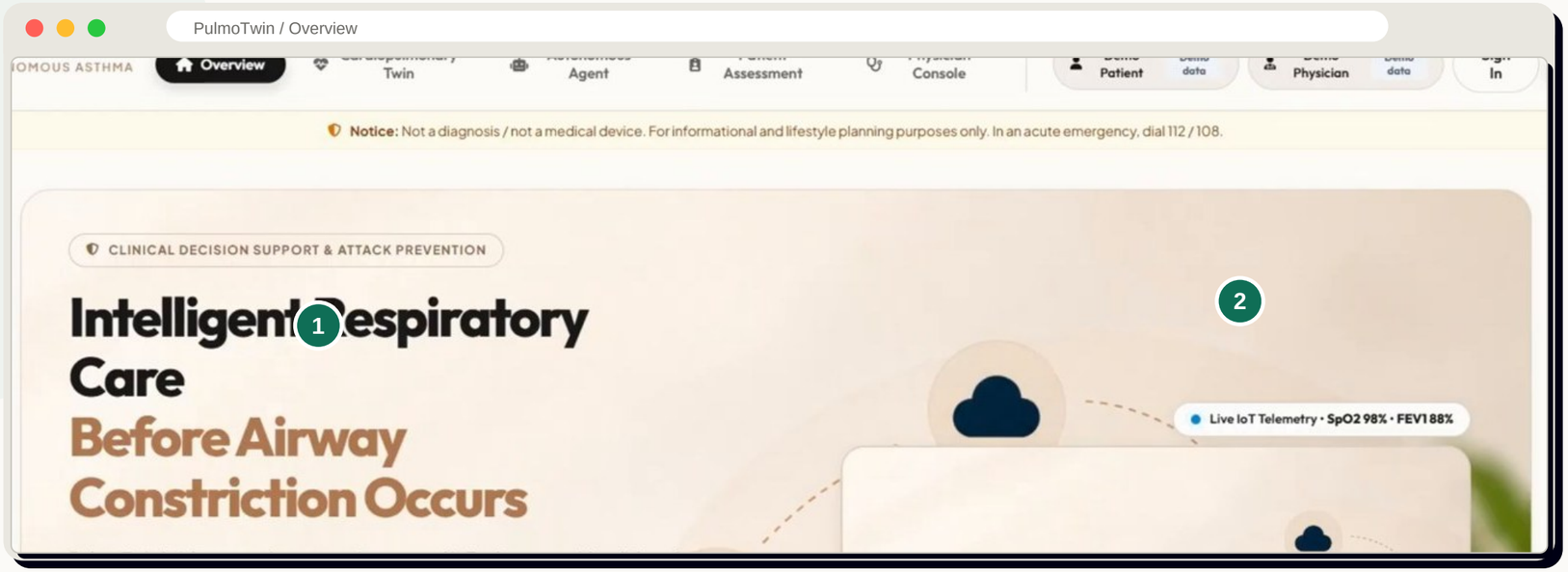
Quality

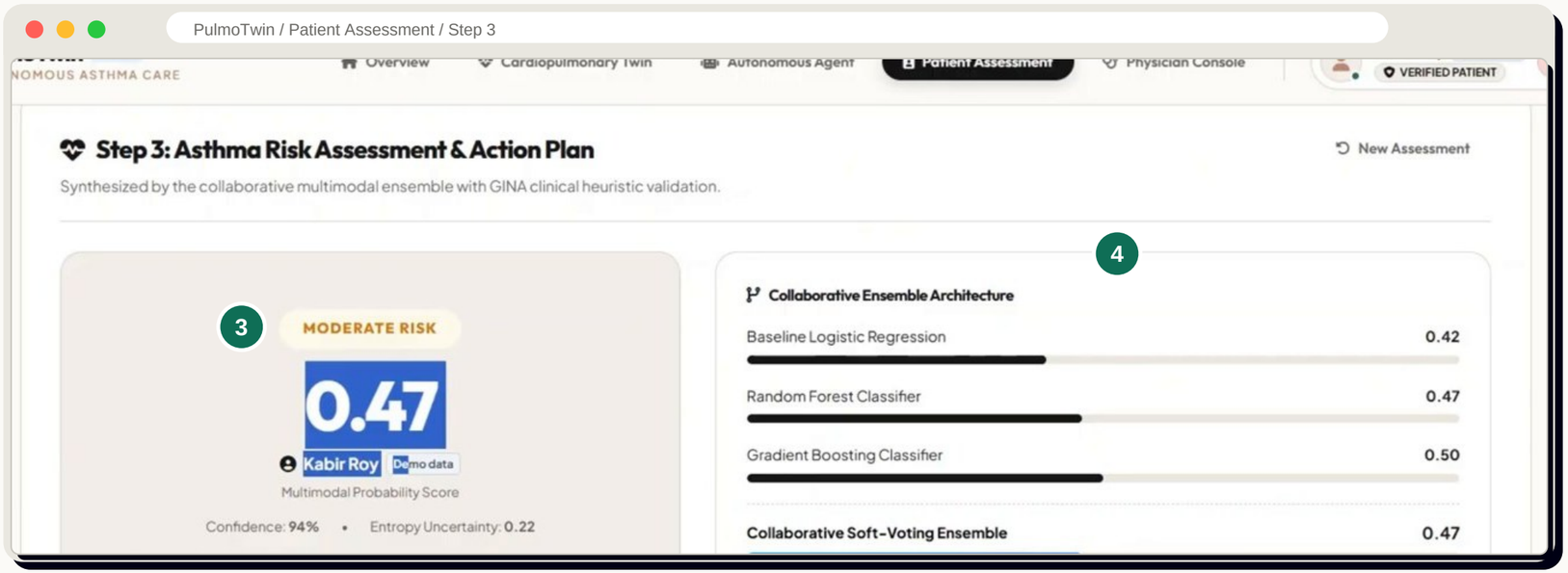
Automated pytest suite for guardrails, agent, twin engine, API and front end. QA and audit reports in the repo.

Deployment

Live GitHub Pages demo with in-browser engine. Render, Vercel, Docker and Procfile for the Flask service.

User interface 1/4: entry and risk assessment





1 One entry point

The overview page leads to the patient app, physician console and assessment, with a live telemetry preview.

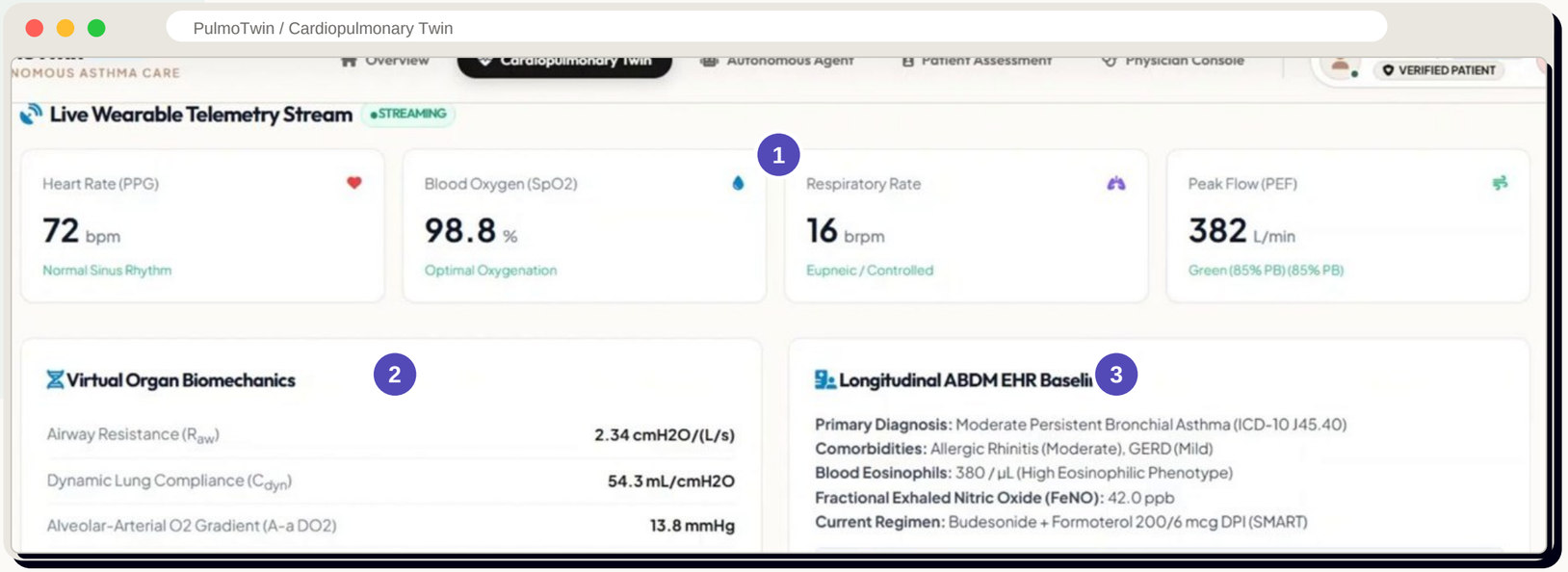
2 Three-step wizard

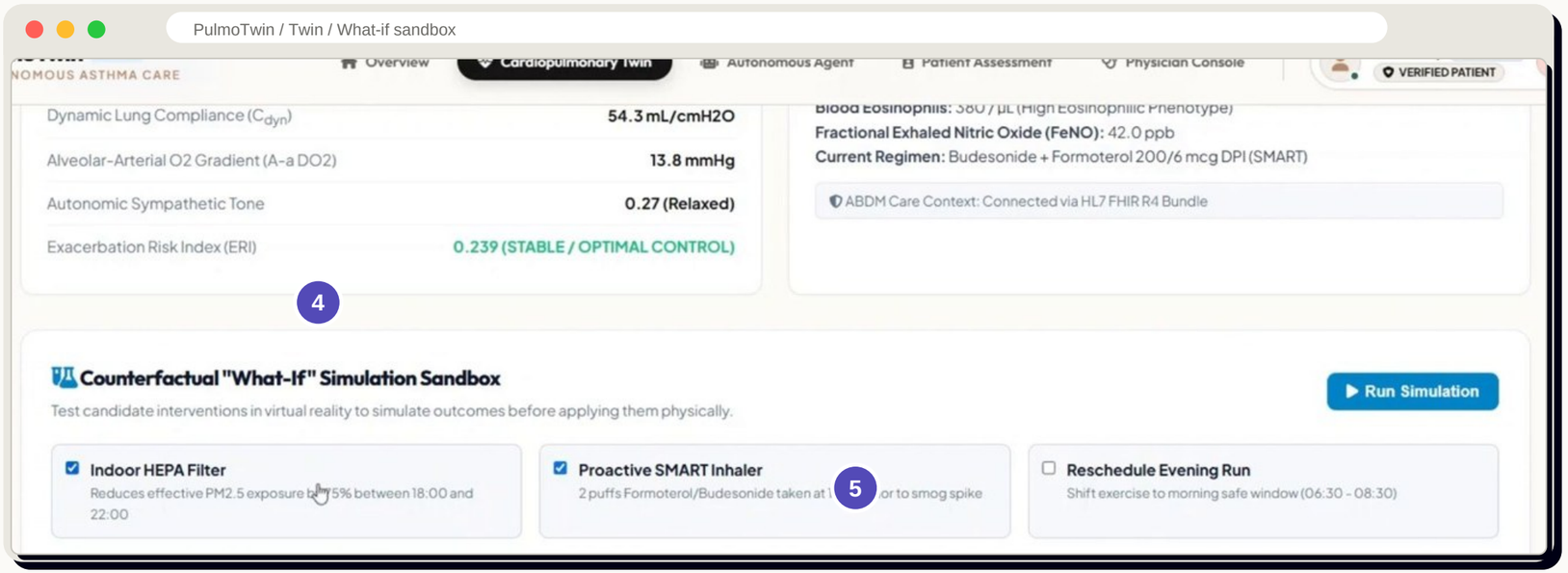
Guideline screening, air-quality sync and a result, so a first-time user is done in about a minute.

3 An explained score

The 0.47 moderate score shows each model's vote, confidence and uncertainty, plus a care plan and SOS.

User interface 2/4: the twin cockpit and what-if sandbox





1 Live replica

Heart rate, SpO2, breathing rate and peak flow stream in; resistance, compliance and oxygen gap are computed.

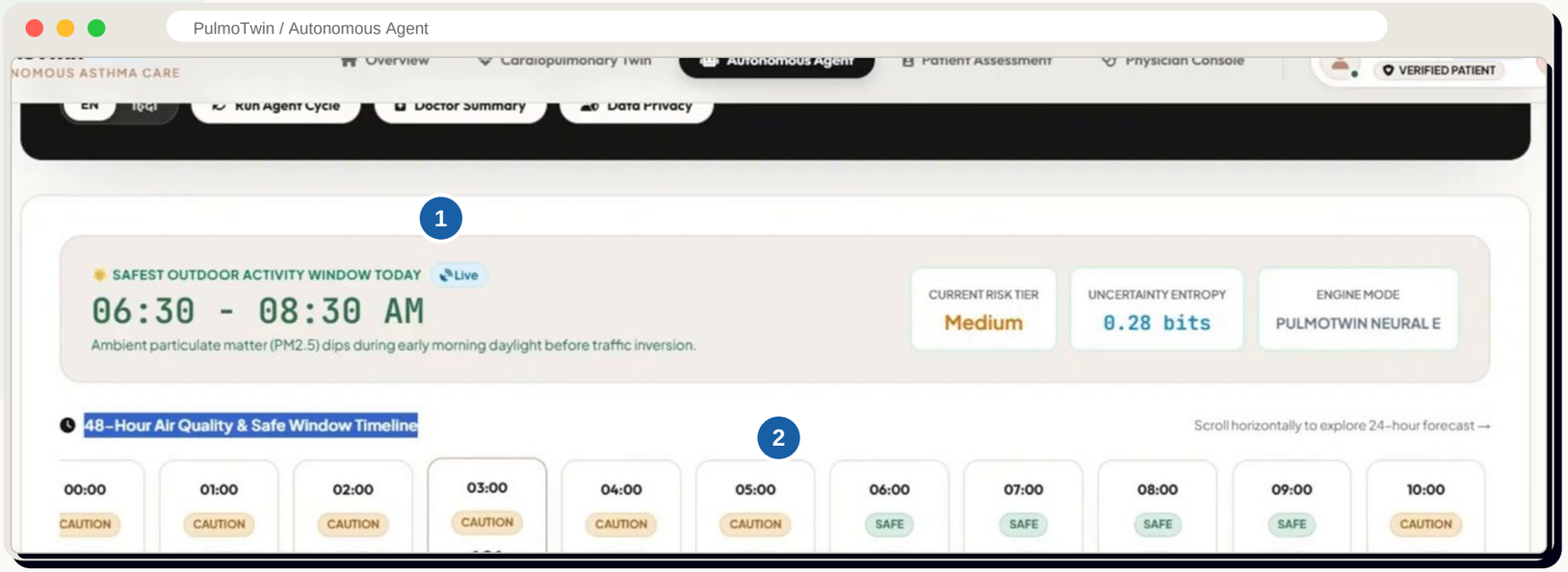
2 Linked to the record

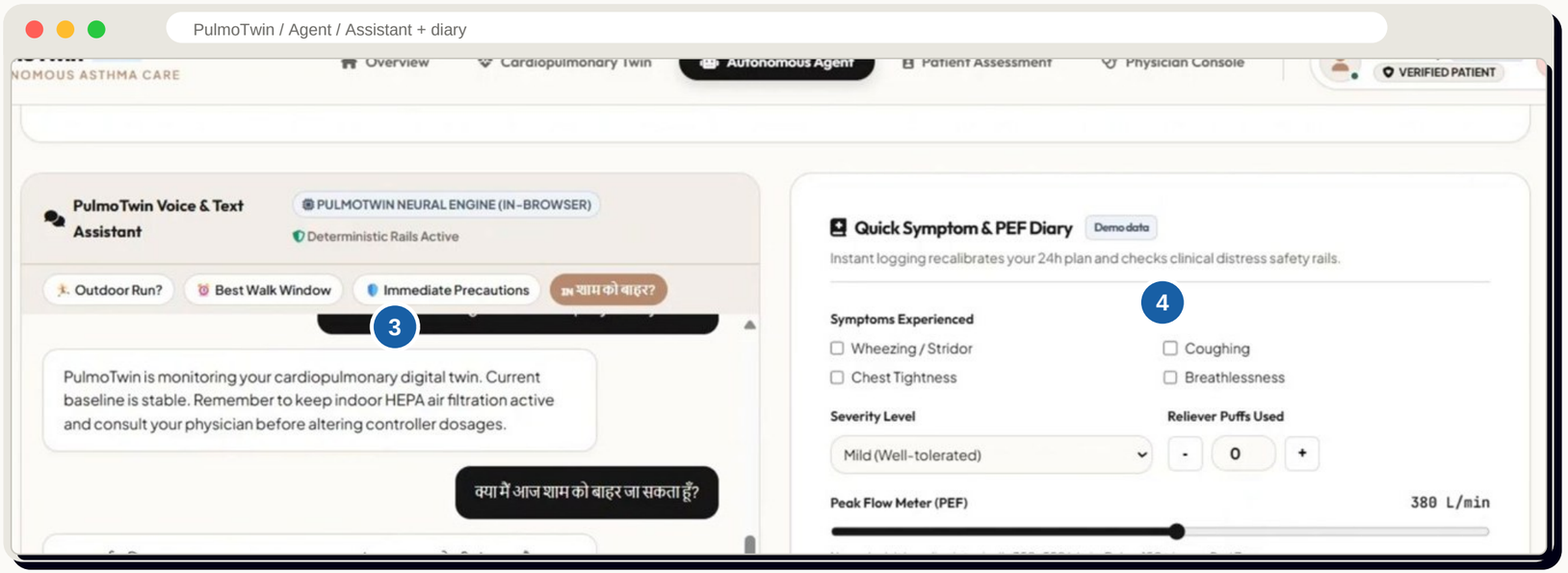
Diagnosis, eosinophils, FeNO and the current regimen come from the ABDM / FHIR baseline.

3 Try before acting

Switch HEPA filter, proactive inhaler and shifted run on or off, then run the simulation.

User interface 3/4: the proactive agent





1 Plans the day

The agent finds the safest outdoor window and lists steps like sealing windows at the evening peak.

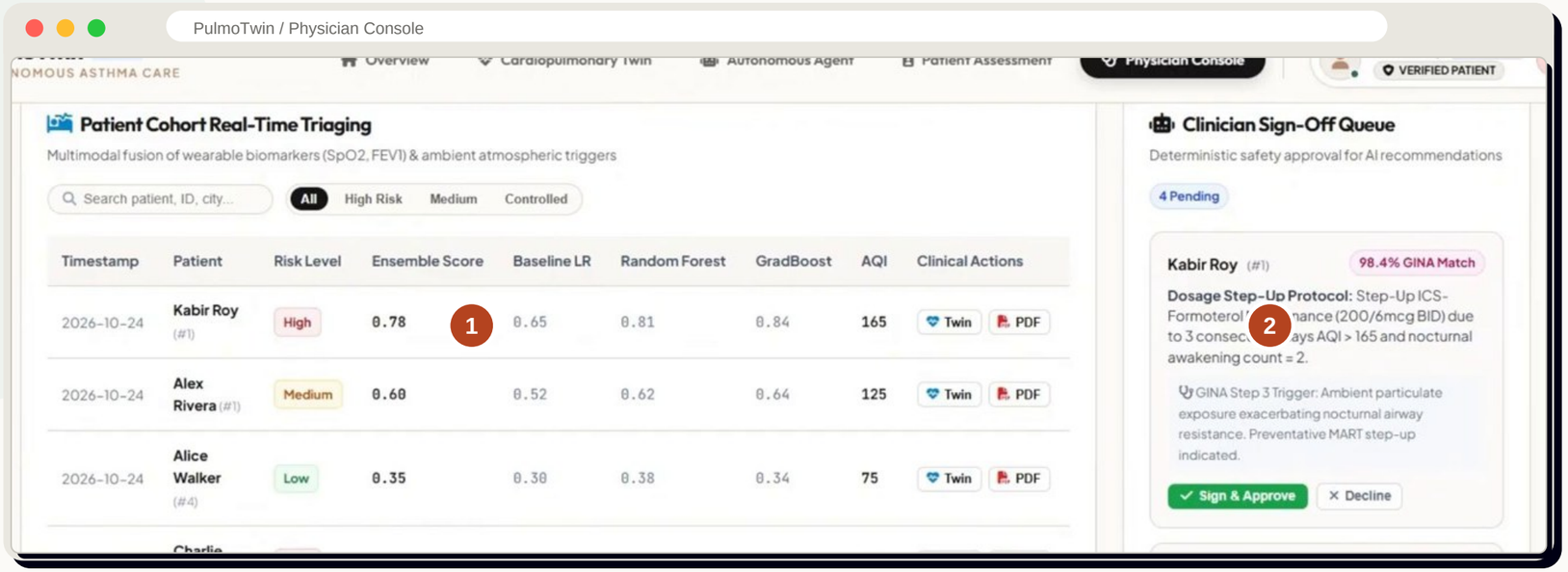
2 Speaks the language

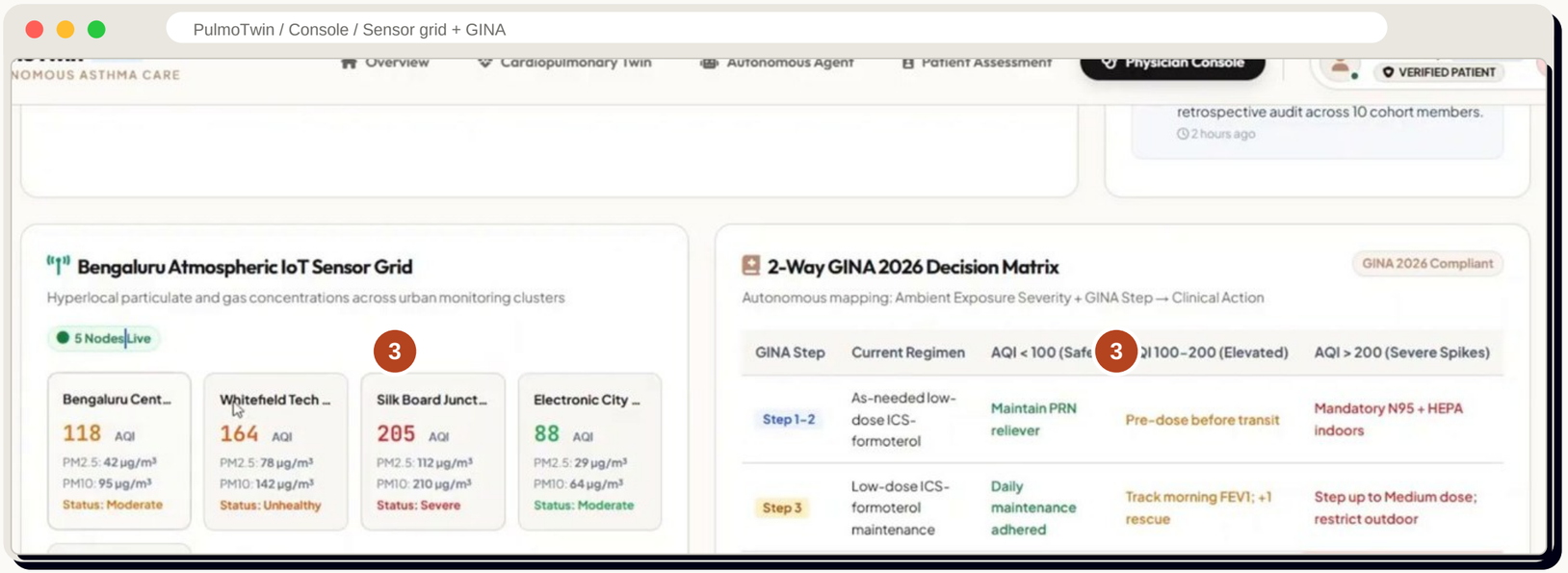
One tap switches the interface and the assistant between English and Hindi, voice or text.

3 Learns from the log

A quick symptom and peak-flow entry recalibrates the plan. Every action waits for approval.

User interface 4/4: the physician console





1 Who first

The triage table ranks patients by risk, with ensemble, LR, RF and GB scores and local AQI.

2 Human in charge

Each AI plan shows its rationale and waits for sign-off or decline from the clinician.

3 Act and document

Quick prescribe issues a GINA-step directive; a 30-day PDF supports the consultation.

System architecture

PulmoTwin | System Architecture

Multimodal cardiopulmonary digital twin for proactive adverse-event forecasting and personalised clinical intervention

Digital Twin Challenge 2026 - Happiest Health Summit
Chronic asthma / cardiopulmonary care, India
Data flows left to right

1 DATA SOURCES
Multimodal inputs

ABDM / HL7 FHIR R4 EHR
digital_twin/ehr_connector.py
- Longitudinal static baseline record
- Spirometry (FEV1, personal-best PEF), FeNO, IgE, blood eosinophils
- Diagnosis (ICD-10), comorbidities, regimen, past exacerbations
- Synthetic data modelled on Synthea / MIMIC-IV, zero PII

Wearable IoT telemetry
digital_twin/wearable_sim.py
- Continuous PPG: heart rate, HRV rMSSD, SpO2, respiratory rate
- Peak flow (PEF) and FEV1 readings
- Smart inhaler actuation over BLE (puff count and timing)
- Simulated sensor stream in the prototype

Atmospheric telemetry
Open-Meteo air quality + weather APIs
- Live GPS sync with 48-hour hourly forecast
- US AQI, PM2.5, PM10, NO2, SO2, O3, CO2
- Temperature and humidity
- City sensor grid feeds the physician view

Patient input
web_ui/index.html
- Bilingual English / Hindi intake, voice and text
- Symptom and PEF diary: wheeze, cough, chest tightness, reliever puffs
- 3-step assessment wizard, manual entry fallback
- Free text is safety-scanned before any model sees it

2 INGESTION AND BACKEND
Fusion, API, storage

Connectors and adapters
ehr_connector.py, wearable_sim.py
- FHIR R4 bundle mapped to the internal patient schema
- Wearable and inhaler events normalised into time series
- Open-Meteo data fetched per GPS location
- data_generator.py builds the synthetic cohort

Preprocessing and fusion
preprocess.py, utils.py
- Cleans and encodes clinical, sensor and air-quality features
- Fuses the three streams into one multimodal feature vector
- Fitted pipeline saved in preprocessing.pkl

Flask REST API
app.py
- Flask 3.1 served by Gunicorn
- /api/digital-twin/* endpoints, including simulate
- Token authentication and IDOR guards
- Serves the patient portal and physician console

Persistence
SQLite via Flask-SQLAlchemy
- Profiles, sensor readings, diary entries, agent plans
- Tamper-evident audit trail of every action
- Stored locally, never sold or shared without consent

3 DIGITAL TWIN ENGINE
Simulate, predict, test

Physiological twin
digital_twin/twin_engine.py
- Airway resistance: Raw = 1.8 / PEF_ratio^1.6
- Lung compliance: Cdyn = 70 x PEF_ratio - FeNO/10
- Alveolar-arterial O2 (A-a DO2) gradient
- Autonomic sympathetic tone from HRV rMSSD
- Exacerbation Risk Index (ERI) from 0 to 1

Multimodal risk ensemble
models.py
- Soft-voting stack: Logistic Regression (w1), Random Forest, 100 trees (w2), Gradient Boosting, 150 stages (w2)
- Accuracy 73.0%, Brier 0.1393 on N=300 held-out
- Shannon-entropy uncertainty rejects low-confidence output

24-hour forward projection
twin_engine.py (ODE kinetics)
- Differential kinetics of airway inflammation
- Forecasts Raw surges and exacerbations 8 to 24 h before onset
- ERI tiers: green controlled, yellow caution, red critical
- Feeds the 48-h safe-window timeline

Counterfactual what-if sandbox
digital_twin/counterfactual.py
- Tests interventions in silico before acting
- Indoor HEPA filter: PM2.5 down 75%, risk down 45.2%
- Proactive SMART inhaler: risk down 66.7%, ERI 0.84 to 0.28
- Shifts evening run to the 06:30-08:30 window

4 SAFETY RAILS
Deterministic, no black box

Acute distress interceptor
input guard, runs first
- 61 distress phrases and 32 physical signs
- English, Hindi and Hinglish, negation-aware
- Bypasses the model: instant SOS, dial 108 / 112
- Also fires when PEF drops below 50% of best

GINA clinical override
GINA Step-5 rules
- Daily symptoms or frequent night waking force High Risk
- High-risk recall rises 72.09% to 82.95% (107 of 129)
- Deliberate bias: a false alarm costs less than a miss

Output sanitizer
post-generation filter
- Regex check on every agent reply
- Blocks dosage changes and diagnostic claims
- Swaps violations for a clinical fallback disclaimer

Zero-trust approval queue
human in the loop
- Agent never sends SMS or edits external systems itself
- Every action stays PENDING until a human approves
- Clinician sign-off queue for physician-side plans

5 INTERFACES AND OUTPUTS
Patient and clinician

Patient portal
web_ui/index.html
- Twin cockpit: live telemetry, organ biomechanics, What-If
- Autonomous agent: 48-h air-quality timeline, voice and text assistant, symptom diary
- Patient assessment: risk score with action plan

Physician console
view-admin
- Cohort triage table with per-model scores
- KPI cards: cohort, sensor nodes, inferences, alerts
- Clinician sign-off queue and Quick Prescribe (GINA)
- Broadcast advisory to the cohort

Reports and privacy
ReportLab 5.0
- 30-day GINA consultation summary PDF
- JSON export for data portability
- Permanent account and health-record erasure
- Aligned with DPDP Act 2023 and GDPR principles

Deployment targets
Pages, Render, Vercel, Docker
- GitHub Pages: in-browser physiology engine
- Render, Vercel, Docker and Procfile for the Flask app
- pytest suite covers guardrails, agent and API

TECHNOLOGY STACK Icons are simplified vector marks, not official brand logos

Python 3.11 core language	Flask 3.1 REST API, Gunicorn	SQLite SQLAlchemy ORM	NumPy, SciPy ODE twin engine	scikit-learn stacking ensemble	ReportLab PDF reports	HL7 FHIR R4 ABDM schemas	Open-Meteo air quality, weather	Docker container, Render	HTML5, ES6 JS web cockpit

PulmoTwin is an investigational decision-support research prototype, not a certified medical device. In an emergency dial 112 / 108.
Source: github.com/Kabirroy12345/Digital_Twin | MIT license | Metrics from README held-out evaluation (N=300)

Sources
EHR, wearables, air, patient input

Backend
Flask API, fusion, SQLite, audit

Twin engine
Physiology, ensemble, projection, what-if

Safety rails
Four checks before any output

Interfaces
Patient app, console, reports

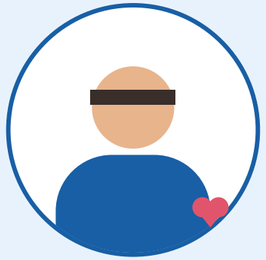
Who we serve



People with asthma

WHO
Adults in polluted Indian cities

WHAT THEY GET
Safe window, early alerts, bilingual help



Caregivers and family

WHO
The people who respond to an attack

WHAT THEY GET
Alerts sent only after the patient approves



Physicians

WHO
Clinicians with short consultations

WHAT THEY GET
Triage, sign-off queue, 30-day PDF



Public-health teams

WHO
Programmes watching air and risk

WHAT THEY GET
Sensor grid and cohort exposure view

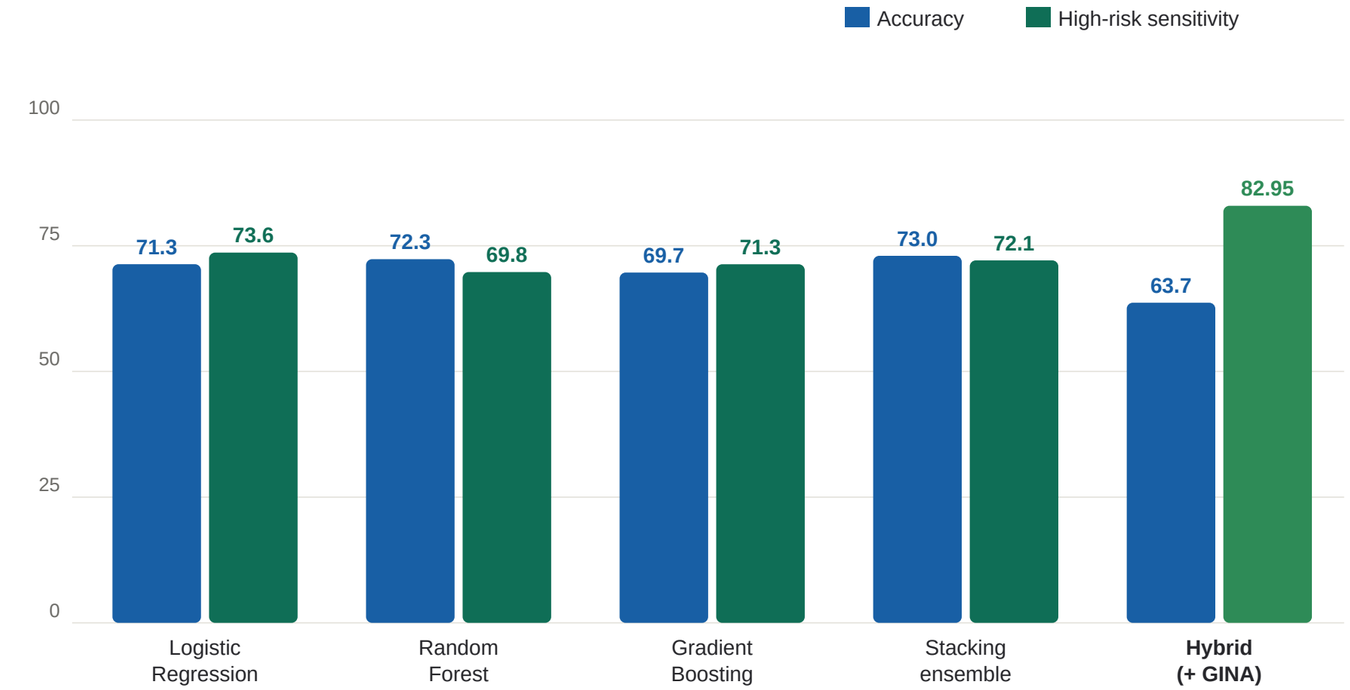
HOW INFORMATION FLOWS BETWEEN THEM

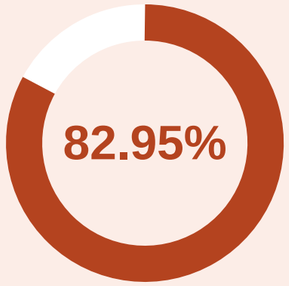


Every step that reaches another person passes through an approval or a clinician, never straight from the AI

Research and results

Accuracy vs high-risk sensitivity on 300 held-out cases (%)





High-risk sensitivity

107 of 129 acute cases caught after the GINA override, up from 72.09%.

A false alarm costs a visit. A miss can cost a life.

0.1393

Brier score

calibration of high-risk probability

73.0%

Stacked accuracy

best single result, pure ML ensemble

HOW WE GOT THERE

Generate cohort
synthetic, FHIR-shaped

Train 3 models
LR, RF, GB

Stack and vote
soft voting

Test on 300
held-out cases

Add GINA rule
measure the trade-off

All results use synthetic data. They show the method, not clinical performance. Next: validation on real, consented cohorts.

Impact: from rescue care to early action

Today

- **12:00** — Feels fine
- **17:00** — Smog builds, unseen
- **20:30** — Symptoms start
- **21:30** — ER or ICU visit

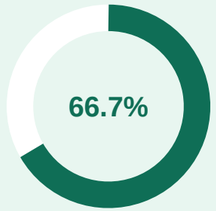
With PulmoTwin

- **12:00** — Twin forecasts a surge
- **13:00** — Alert, 8 hours ahead
- **17:00** — HEPA + inhaler dose
- **21:30** — Stable, green

Timeline times are illustrative; lead time of 8 to 24 h is the project's forecast window.

Patients

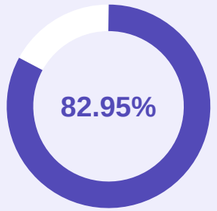
Plan the day, dose early, get help fast



simulated risk cut

Doctors

30 days of objective data, ranked cohort



high-risk sensitivity

Health system

Fewer avoidable ER and ICU episodes



forecast lead time

Figures come from simulation and synthetic data. Real-world outcomes need clinical validation.

How PulmoTwin is different

	Typical AQI or symptom app	Static risk score	PulmoTwin
When it speaks	After symptoms start	At the clinic visit	**8 to 24 h before onset**
What it knows	Generic area air quality	A few questionnaire answers	**Record + body + air, per person**
Data fused	One source	One snapshot	**Three live streams**
Try before acting	No	No	**What-if sandbox**
Safety	Disclaimers	Not applicable	**Rules in code, human approval**
Language	Mostly English	Mostly English	**English and Hindi, voice**
Doctor link	Rare	Manual	**Triage, sign-off, PDF**

The first two columns describe common approaches in general, not specific products.

Key features and safeguards

Live air quality
GPS 48-hour PM2.5, NO2, SO2, O3

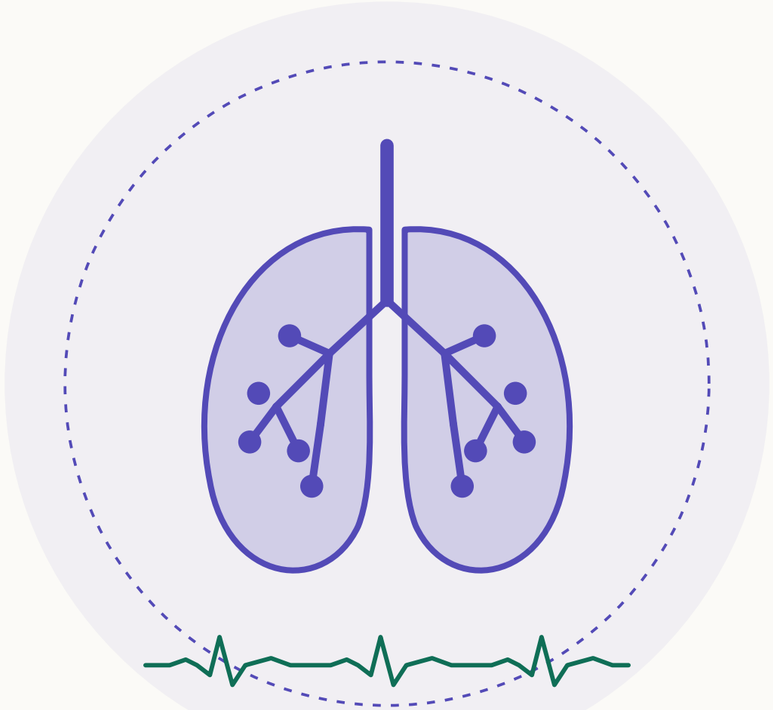
Wearable stream
HR, HRV, SpO2, breathing, peak flow

Physiology twin
Resistance, compliance, O2 gap

24-hour forecast
Risk index hour by hour

What-if sandbox
HEPA, inhaler timing, shifted run

Proactive agent
Safe window and daily plan



PulmoTwin

teal data in · purple twin and AI · blue interfaces · coral safety

Bilingual assistant
English and Hindi, voice and text

Physician console
Triage, sign-off, quick prescribe

Distress bypass
Hindi, English, Hinglish to 108 / 112

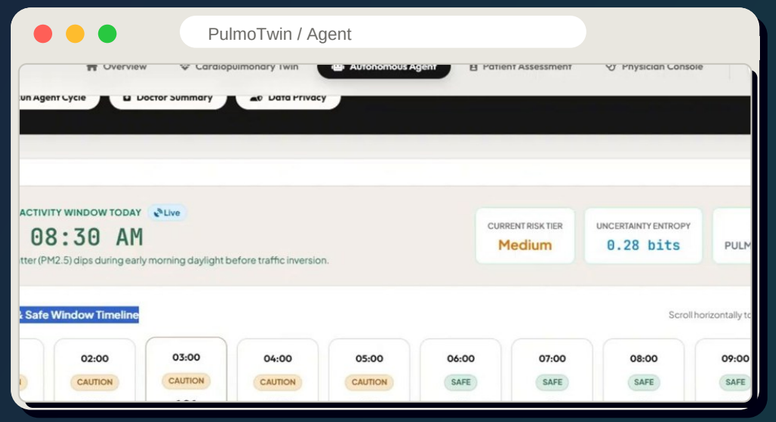
Approval queue
Nothing sent without a human yes

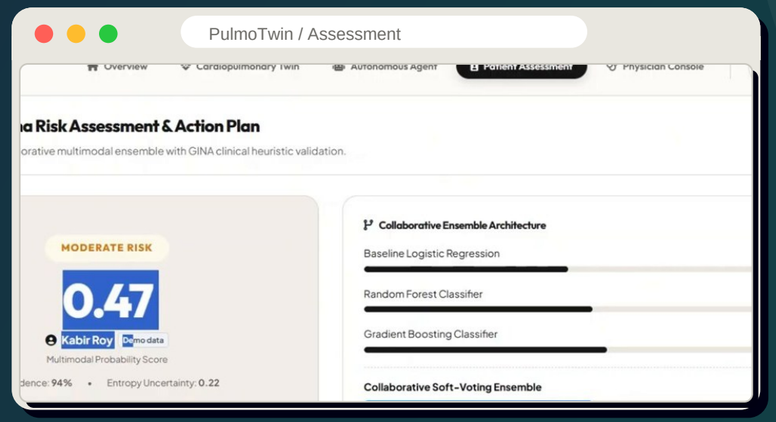
Output filter
Blocks dosing changes, diagnoses

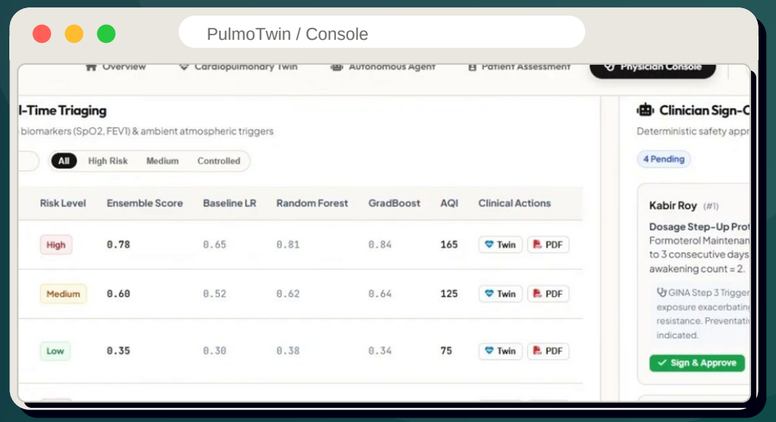
Privacy and audit
Export, erase, audit trail

Thank you

PulmoTwin: see the attack coming, and act before it arrives.







OPEN THE PROJECT (every box is a clickable link)

Live interactive demo
kabirroy12345.github.io/Digital_Twin

Source code on GitHub
github.com/Kabirroy12345/Digital_Twin

Full demo video on YouTube
youtube.com/watch?v=i7NJstq8_ZE

Architecture diagram (PDF)
github.com/Kabirroy12345/Digital_Twin/docs

Team PulmoTwin Digital Twin Challenge 2026 | Happiest Health Summit, Bengaluru | Research prototype, not a medical device. Emergency: 112 / 108